

notebook-philippine-agriculture-reddit-analysis

June 8, 2026

1 CSCI 161.03 - Introduction to Social Computing

Author: BINWAG, Louis III G. (github/louis-uwie)

2 I. Introduction

As part of the major requirement for **CSCI 161.03 - Introduction to Social Computing**, the author,

In **r/Philippines**, users tend to interact about on-going controversial issues in the Philippines. This ranges from the most recent **Nepo Babies** to rather old but still very relevant and recurring, **Agricultural Prices**.

What I want to find out are:

1. What are the common underlying themes in topics concerning **Agriculture**?
2. What **dominant emotions** emerge in these online conversations?
3. What are common indications?

3 II. About the Dataset

Please note that there is a great chance of including other social media platforms as well due to rate-limits of Reddit API Scraping.

Currently, there is 248 Rows, and 8 Columns.

Column	Definition	Sample
post_title	Title of the Reddit post	<i>"Why is Python so popular in data science?"</i>
post_url	Direct link to the Reddit post	https://www.reddit.com/r/datascience/comments/abc123/
post_body	Main body text of the Reddit post (if provided)	<i>"I've noticed a lot of job postings require Python experience..."</i>
post_author	Username of the Reddit post's author	u/data_guru
post_score	Upvotes minus downvotes for the post	1542
comment	Text of the Reddit comment	<i>"Python has great libraries like pandas and scikit-learn."</i>
comment_author	Username of the Reddit comment's author	u/ml_enthusiast

Column	Definition	Sample
comment_score	Upvotes minus downvotes for the comment	237

4 III. Data Collection

Data collection will primarily utilize Reddit's PRAW (API) for datascraping. Connecting to Google drive for ease of use since the author will be using google collab for a more streamline and 'smoother' project.

```
[1]: %pip install praw numpy nltk pandas
```

Collecting praw

Using cached praw-7.8.1-py3-none-any.whl.metadata (9.4 kB)

Requirement already satisfied: numpy in

/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-packages (2.3.3)

Requirement already satisfied: nltk in

/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-packages (3.9.2)

Requirement already satisfied: pandas in

/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-packages (2.3.3)

Collecting prawcore<3,>=2.4 (from praw)

Using cached prawcore-2.4.0-py3-none-any.whl.metadata (5.0 kB)

Collecting update_checker>=0.18 (from praw)

Using cached update_checker-0.18.0-py3-none-any.whl.metadata (2.3 kB)

Requirement already satisfied: websocket-client>=0.54.0 in

/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-packages (from praw) (1.9.0)

Requirement already satisfied: requests<3.0,>=2.6.0 in

/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-packages (from prawcore<3,>=2.4->praw) (2.32.5)

Requirement already satisfied: charset_normalizer<4,>=2 in

/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-packages (from requests<3.0,>=2.6.0->prawcore<3,>=2.4->praw) (3.4.3)

Requirement already satisfied: idna<4,>=2.5 in

/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-packages (from requests<3.0,>=2.6.0->prawcore<3,>=2.4->praw) (3.10)

Requirement already satisfied: urllib3<3,>=1.21.1 in

/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-packages (from requests<3.0,>=2.6.0->prawcore<3,>=2.4->praw) (2.3.0)

Requirement already satisfied: certifi>=2017.4.17 in

/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-packages (from requests<3.0,>=2.6.0->prawcore<3,>=2.4->praw) (2025.10.5)

Requirement already satisfied: click in

/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-

```

packages (from nltk) (8.3.0)
Requirement already satisfied: joblib in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (from nltk) (1.5.2)
Requirement already satisfied: regex>=2021.8.3 in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (from nltk) (2025.9.18)
Requirement already satisfied: tqdm in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (from nltk) (4.67.1)
Requirement already satisfied: python-dateutil>=2.8.2 in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Using cached praw-7.8.1-py3-none-any.whl (189 kB)
Using cached prawcore-2.4.0-py3-none-any.whl (17 kB)
Using cached update_checker-0.18.0-py3-none-any.whl (7.0 kB)
Installing collected packages: update_checker, prawcore, praw

```

3/3

```
[praw][32m2/3 [praw]
```

Successfully installed praw-7.8.1 prawcore-2.4.0 update_checker-0.18.0

[notice] A new release of pip is
available: 25.2 -> 25.3

[notice] To update, run:

```
pip install --upgrade pip
```

Note: you may need to restart the kernel to use updated packages.

```

[2]: # from dotenv import load_dotenv
      # import os

      # load_dotenv("/content/drive/CSCI 161/Project/redditdatascape.env",
      ↪override=True)

      # CLIENT_ID = os.getenv("CLIENT_ID")
      # CLIENT_SECRET = os.getenv("CLIENT_SECRET")

      # print("CLIENT_ID:", CLIENT_ID)
      # print("CLIENT_SECRET:", CLIENT_SECRET)

```

```
[3]: import pandas as pd
import numpy as np
import praw
import pandas as pd
import os
```

```
[ ]: reddit = praw.Reddit(
    client_id="*****",
    client_secret="*****",
    user_agent="script:reddit_scraper:v1.0"
)

print("Read-only mode:", reddit.read_only)
```

Read-only mode: True

```
[5]: subreddit = reddit.subreddit("Philippines")

posts_data = []

# Fetch more posts (adjust as needed)
for post in subreddit.search("Agriculture", limit=500):
    body = post.selftext if post.selftext else "[No text - maybe image/link_
↳post]"

    posts_data.append({
        "post_title": post.title,
        "post_url": "https://reddit.com" + post.permlink,
        "comment": None,
        "comment_author": None,
        "comment_score": None,
        "post_body": body,
        "post_score": post.score,
        "post_author": post.author.name if post.author else "[deleted]",
        "created_utc": pd.to_datetime(post.created_utc, unit="s")
    })

# Fetch comments
post.comments.replace_more(limit=0)
for comment in post.comments.list():
    posts_data.append({
        "post_title": post.title,
        "post_url": "https://reddit.com" + post.permlink,
        "comment": comment.body,
        "comment_author": comment.author.name if comment.author else_
↳"[deleted]",
        "comment_score": comment.score,
```

```

        "post_body": None,
        "post_score": None,
        "post_author": None,
        "created_utc": pd.to_datetime(comment.created_utc, unit="s")
    })

# Save locally
df = pd.DataFrame(posts_data)

output_dir = "/Users/uwie/Documents/_Github/CSCI161-SocialComputing/Binwag -_
↳PhilippineRedditAgricultureAnalysis/data"
os.makedirs(output_dir, exist_ok=True)

output_path = os.path.join(output_dir, "reddit_data.csv")
df.to_csv(output_path, index=False)

```

```

[6]: reddit_collected = pd.read_csv(
    "/Users/uwie/Documents/_Github/CSCI161-SocialComputing/Binwag -_
↳PhilippineRedditAgricultureAnalysis/data/reddit_data.csv"
)
reddit_collected.tail()

```

```

[6]:
                                     post_title \
13762  121 years ago on July 2, 1899, the Manila Agri...
13763  World Soil Day: How the SoilMate App is Tackli...
13764  Benguet State University - BS Agriculture Feed...
13765  Benguet State University - BS Agriculture Feed...
13766  Benguet State University - BS Agriculture Feed...

                                     post_url \
13762  https://reddit.com/r/Philippines/comments/hjkb...
13763  https://reddit.com/r/Philippines/comments/k9jf...
13764  https://reddit.com/r/Philippines/comments/gc5m...
13765  https://reddit.com/r/Philippines/comments/gc5m...
13766  https://reddit.com/r/Philippines/comments/gc5m...

                                     comment  comment_author \
13762          Nice. I see what you did there.  PhilippineLeadX
13763                                     NaN                  NaN
13764                                     NaN                  NaN
13765          Dude can i join you?  readytraderone
13766  Maganda naman. Hindi malaki ang bayarin.  fmp321

                                     comment_score  post_body \
13762          3.0                                     NaN
13763          NaN  [No text - maybe image/link post]
13764          NaN  Hello guys!\n\nI'm currently 29 years old and ...

```

13765	1.0	NaN
13766	1.0	NaN

	post_score	post_author	created_utc
13762	NaN	NaN	2020-07-02 11:34:00
13763	2.0	cottage_ogre	2020-12-09 02:57:18
13764	3.0	agriwannabe	2020-05-02 12:26:35
13765	NaN	NaN	2020-05-02 16:29:37
13766	NaN	NaN	2020-05-02 16:58:55

Rappler - News Article

```
[7]: import requests
from bs4 import BeautifulSoup
import pandas as pd
import time

# =====
# CONFIGURATION
# =====

BASE_TOPIC_URL = "https://www.rappler.com/topic/agriculture/page/{}/"
START_PAGE = 1
END_PAGE = 5 # adjust after confirming how many pages exist
OUTPUT_FILE = "rappler_agriculture.csv"

# Initialize DataFrame
corpus = pd.DataFrame(columns=['title', 'link', 'date_published', 'author', 'text'])

# =====
# SCRAPER FUNCTIONS
# =====

def get_article_details(article_url):
    """Extract title, date, author, and text from an article."""
    try:
        res = requests.get(article_url, timeout=10)
        res.raise_for_status()
        soup = BeautifulSoup(res.text, "html.parser")

        # Title
        title_tag = soup.find("h1")
        title = title_tag.get_text(strip=True) if title_tag else None
```

```

    # Date
    date_tag = soup.find("time")
    date_published = date_tag.get("datetime") if date_tag else None

    # Author
    author_tag = soup.find("a", rel="author") or soup.find("span",
↪class_="author__name")
    author = author_tag.get_text(strip=True) if author_tag else None

    # Article text
    paragraphs = soup.select("article p")
    text = " ".join(p.get_text(strip=True) for p in paragraphs) if
↪paragraphs else None

    return [title, article_url, date_published, author, text]

except Exception as e:
    print(f"Error scraping article {article_url}: {e}")
    return [None, article_url, None, None, None]

def scrape_topic_page(page_number):
    """Scrapes one page of the Agriculture topic."""
    url = BASE_TOPIC_URL.format(page_number)
    print(f"Scraping topic page {page_number}: {url}")

    try:
        res = requests.get(url, timeout=10)
        res.raise_for_status()
        soup = BeautifulSoup(res.text, "html.parser")
    except Exception as e:
        print(f"Failed to load topic page {page_number}: {e}")
        return []

    articles_data = []

    # On archive pages, each article link is inside <h2><a href=...>
    article_links = [a['href'] for a in soup.select("h2 > a[href]")]
    if not article_links:
        print(f"No articles found on page {page_number}.")
        return []

    for link in article_links:
        if not link.startswith("http"):
            link = "https://www.rappler.com" + link

        article_details = get_article_details(link)

```

```

        articles_data.append(article_details)
        print(f"Scraped: {article_details[0]}")
        time.sleep(1) # polite delay

    return articles_data

# =====
# MAIN LOOP
# =====

all_data = []

for page in range(START_PAGE, END_PAGE + 1):
    page_data = scrape_topic_page(page)
    if not page_data:
        print(f"Stopping at page {page} (no more articles).")
        break
    all_data.extend(page_data)

# Combine into DataFrame
corpus = pd.DataFrame(all_data, columns=['title', 'link', 'date_published', 'author', 'text'])

# Save to CSV
corpus.to_csv(OUTPUT_FILE, index=False, encoding='utf-8-sig')
print(f"Scraping complete. {len(corpus)} articles saved to {OUTPUT_FILE}.")

```

Scraping topic page 1: <https://www.rappler.com/topic/agriculture/page/1/>
 Scraped: Apl.de.Ap extends his impact from stage to soil
 Scraped: What is Japan's Matsusaka beef, and why is it so rare?
 Scraped: Arms to farms: Maguindanao del Sur's agriculture roadmap for former rebels
 Scraped: Chinese smugglers 'operate freely' with help of PH gov't officials - Pangilinan
 Scraped: Bounty Fresh aims to strengthen operations amid growing appetite for poultry
 Scraped: Century Pacific buys another coconut facility. What's next for its food business?
 Scraped: [Ask the Tax Whiz] How do Filipinos benefit from a higher excise tax on sugary drinks?
 Scraped: DA to take over farm-to-market road projects from DPWH
 Scraped: Zaldivar-linked firm ranks No. 3 in 2024 farm-to-market road projects
 Scraping topic page 2: <https://www.rappler.com/topic/agriculture/page/2/>
 Scraped: Trump's tariff war is ASEAN ministers' lunchtime talk: 'Have you closed?'
 Scraped: Cheng family, owner of Bounty Fresh, now has 200 restaurants via The Bistro Group

Scraped: NFA needs P3 billion for emergency palay procurement

Scraped: DAR eyes P17.4-billion budget for 2026

Scraped: South Korean president stops loan for DAR's 'PBBM Bridges' project over corruption concerns

Scraped: Pangilinan wants consignees, brokers of smuggled goods blacklisted

Scraped: 'Raket 'yun': Disputes over location delays PH's first border inspection facility

Scraped: LIVESTREAM: Senate panel probes high food prices

Scraped: Marcos sinuyo mga lokal at banyagang negosyo: Mamuhunan kayo sa agrikultura ng Pilipinas

Scraped: Marcos woos local, foreign businesses: Invest in Philippine agriculture

Scraping topic page 3: <https://www.rappler.com/topic/agriculture/page/3/>

Scraped: Beyond benteng bigas: Advocates slam Marcos' weak agri efforts ahead of SONA 2025

Scraped: [Time Trowel] The Ifugao Rice Terraces are not 2,000 years old

Scraped: [OPINION] Why the Philippine Agriculturists Act gets rural development wrong

Scraped: Pangilinan eyes Senate agri committee. Will Villar inherit post from her mother?

Scraped: WATCH: Marcos' P20/kilo rice promise fulfilled? Yes and no.

Scraped: Group blames cheap imported rice for drop in palay prices

Scraped: [Time Trowel] Ube: History you can eat

Scraped: Gov't pilots rice crop insurance using satellite data

Scraped: ADB to invest \$500M to cut down PH rice sector emissions

Scraped: If Comelec allows, DA to launch P20/kilo rice program on May 1

Scraping topic page 4: <https://www.rappler.com/topic/agriculture/page/4/>

Scraped: [Good Business] Puso ng saging, puso ng tao: How loob shapes kapwa at Villa Socorro Farm

Scraped: WATCH: Why the Villar family's real estate interests are an election issue

Scraped: More Veggies Please? Metro Pacific eyes 10 more vegetable greenhouse facilities

Scraped: Palace, DA push Congress to revisit Rice Tariffication Law

Scraped: DA urges NBI to probe claims farmers took their lives due to low palay prices

Scraped: PH records lower rice imports in Q1 2025

Scraped: [Good Business] Island Harvest Farm's kaloob: A gift that keeps on giving

Scraped: Current Kanlaon crisis hurting La Castellana's P50-million banana industry

Scraped: DA sees low compliance on pork price cap in NCR

Scraped: P2.83 billion worth of smuggled agricultural goods seized in 2024 - DA

Scraping topic page 5: <https://www.rappler.com/topic/agriculture/page/5/>

Scraped: Gov't urges LGUs to hasten NFA rice pullout

Scraped: DA to lower rice max SRP to P49/kilo by March

Scraped: Arctic doomsday seed vault gets more than 14,000 new samples

Scraped: San Juan City first to sell NFA rice after food emergency is declared

Scraped: [Good Business] A call to future senators to rethink Philippine

agriculture

Scraped: DA to release NFA rice in Metro Manila, Camarines Sur

Scraped: Billionaire Eduardo Soriano, Anscor vice chairman, dies

Scraped: No pork prices above P400/kilo? DA to set max SRP by March

Scraped: Kanlaon eruptions leave Negros Island Region with nearly P1B in farm losses

Scraped: DA okays onion importation to avoid repeat of 2022 crisis, farmers question timing

Scraping complete. 49 articles saved to rappler_agriculture.csv.

5 Data Preprocessing

```
[8]: # conda activate "F:\_GitHub\CSCI161-SocialComputing\Binwag -  
      ↪PhilippineRedditAgricultureAnalysis\env"  
      # OR go into the directory above, and conda activate  
      # conda install pytorch torchvision torchaudio pytorch-cuda=12.1 -c pytorch -c  
      ↪nvidia  
  
      # conda activate ./env  
      # jupyter notebook
```

```
[9]: import os  
      import sys  
      import subprocess  
  
      print("Checking GPU and environment...")  
  
      conda_env = os.environ.get("CONDA_PREFIX", None)  
      print(f"Conda environment: {conda_env if conda_env else 'Not in Conda'}")  
  
      try:  
          result = subprocess.run(["nvidia-smi"], stdout=subprocess.PIPE,  
                                  ↪stderr=subprocess.PIPE, text=True)  
          if result.returncode == 0:  
              print("NVIDIA GPU detected.")  
              print(result.stdout.split("\n")[2])  
          else:  
              print("No NVIDIA GPU found or drivers missing.")  
      except FileNotFoundError:  
          print("'nvidia-smi' not found - NVIDIA drivers may not be installed.")  
  
      try:  
          import torch  
          print("PyTorch already installed.")  
      except ImportError:  
          print("Installing PyTorch with CUDA 12.1 support...")
```

```

subprocess.check_call([
    sys.executable, "-m", "pip", "install",
    "torch", "torchvision", "torchaudio",
    "--extra-index-url", "https://download.pytorch.org/whl/cu121"
])
import torch

try:
    import torch
    print("PyTorch CUDA available:", torch.cuda.is_available())
    if torch.cuda.is_available():
        print("Using GPU:", torch.cuda.get_device_name(0))
    else:
        print("CUDA not available - running on CPU.")
except Exception as e:
    print("PyTorch check failed:", e)
    print("Trying TensorFlow GPU check instead...")
    subprocess.check_call([sys.executable, "-m", "pip", "install",
        ↪ "tensorflow[and-cuda]"])
    import tensorflow as tf
    gpus = tf.config.list_physical_devices('GPU')
    print("TensorFlow GPUs detected:" if gpus else "TensorFlow found no GPU.",
        ↪ gpus)

print("Setup complete.")

```

Checking GPU and environment...

Conda environment: Not in Conda

'nvidia-smi' not found - NVIDIA drivers may not be installed.

Installing PyTorch with CUDA 12.1 support...

Looking in indexes: <https://pypi.org/simple>,

<https://download.pytorch.org/whl/cu121>

ERROR: Could not find a version that satisfies the requirement torch (from
versions: none)

[notice] A new release of pip is

available: 25.2 -> 25.3

[notice] To update, run:

pip install --upgrade pip

ERROR: No matching distribution found for torch

```

-----
ModuleNotFoundError
Cell In[9], line 21

```

Traceback (most recent call last)

```

20 try:
---> 21     import torch
22     print("PyTorch already installed.")

```

ModuleNotFoundError: No module named 'torch'

During handling of the above exception, another exception occurred:

CalledProcessError Traceback (most recent call last)

Cell In[9], line 25

```

23 except ImportError:
24     print("Installing PyTorch with CUDA 12.1 support...")
---> 25     subprocess.check_call([
26         sys.executable,
27         'pip', 'install', 'torch',
28         'torchvision', 'torchaudio',
29     ])
30     import torch
32 try:

```

```

File /usr/local/Cellar/python@3.13/3.13.3/Frameworks/Python.framework/Versions/.
↳13/lib/python3.13/subprocess.py:419, in check_call(*popenargs, **kwargs)
417     if cmd is None:
418         cmd = popenargs[0]
--> 419     raise CalledProcessError(retcode, cmd)
420 return 0

```

```

CalledProcessError: Command '['/Users/uwie/Documents/_Github/
↳CSCI161-SocialComputing/.venv/bin/python', '-m', 'pip', 'install', 'torch',
↳'torchvision', 'torchaudio', '--extra-index-url', 'https://download.pytorch.
↳org/whl/cu121']' returned non-zero exit status 1.

```

```

[ ]: import pandas as pd
import os

print(f"Current directory: {os.getcwd()}")

reddit_raw = pd.read_csv(r"data/Reddit/reddit_data.csv")
rappler_raw = pd.read_csv(r"data/News/rappler_agriculture.csv")

reddit_raw.info()
reddit_raw.tail()

[ ]: rappler_raw.info()
rappler_raw.tail()

```

Cleaning Reddit Data

```
[ ]: import pandas as pd
import re

reddit_clean = reddit_raw.drop(columns=['comment_author',
↳ 'post_author', 'post_score', 'comment_score'])

reddit_clean = reddit_clean.dropna(subset=['post_title', 'comment'], how='all')

# Combine title + body into one field (if body exists)
reddit_clean['post_full'] = reddit_clean.apply(
    lambda row: f"{row['post_title']} {row['post_body']}" if pd.
↳ notnull(row['post_body']) else row['post_title'],
    axis=1
)

# Text normalization
def clean_text(text):
    if pd.isna(text):
        return ""
    text = re.sub(r"http\S+|www\S+", "", text)      # remove URLs
    text = re.sub(r"<.*?>", "", text)              # remove HTML tags
    text = re.sub(r"[\r\n]+", " ", text)           # newlines → space
    text = re.sub(r"\s+", " ", text).strip()        # normalize whitespace
    text = text.lower()                             # lowercase
    text = text.replace("[deleted]", "").replace("[removed]", "")
    return text.strip()

reddit_clean['post_full'] = reddit_clean['post_full'].apply(clean_text)
reddit_clean['comment'] = reddit_clean['comment'].apply(clean_text)

reddit_clean = reddit_clean[(reddit_clean['post_full'].str.len() > 0) |
↳ (reddit_clean['comment'].str.len() > 0)]

reddit_clean = reddit_clean.drop_duplicates(subset=['post_url', 'comment'])

reddit_clean = reddit_clean.reset_index(drop=True)

reddit_clean.to_csv("data/Reddit/reddit_clean.csv", index=False,
↳ encoding='utf-8-sig')

print(f"Cleaned {len(reddit_clean)} rows saved to reddit_clean.csv")
reddit_clean.tail(3)
```

Reddit is cleaned by normalizing text, duplicates, and removing unused columns for this study.

```
[ ]: rappler_clean = rappler_raw.copy()

# Drop the 'author' column
rappler_clean = rappler_clean.drop(columns=['author'])

# Text normalization function
def clean_text(text):
    if pd.isna(text):
        return ""
    text = re.sub(r"http\S+|www\S+", "", text)      # remove URLs
    text = re.sub(r"<.*?>", "", text)              # remove HTML tags
    text = re.sub(r"[\r\n]+", " ", text)           # newlines → space
    text = re.sub(r"\s+", " ", text).strip()        # normalize whitespace
    text = text.lower()                             # lowercase
    text = text.replace("[deleted]", "").replace("[removed]", "")
    return text.strip()

rappler_clean['cleaned_text'] = rappler_clean['text'].apply(clean_text)
rappler_clean.to_csv(r"data/News/rappler_clean.csv", index=False)

rappler_clean.tail()
```

Final overview

```
[ ]: rappler_clean.info()
rappler_clean.isna().sum()
```

```
[ ]: rappler_clean[['text', 'cleaned_text']].tail()
```

6 EDA

reddit_clean and rappler_clean will be used.

Basic dataset overview

```
[ ]: import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
from wordcloud import WordCloud
from wordcloud import STOPWORDS

reddit_clean = pd.read_csv(r"../data/Reddit/reddit_clean.csv")
rappler_clean = pd.read_csv(r"../data/News/rappler_clean.csv")

reddit_clean.info()
```

```
rappler_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12730 entries, 0 to 12729
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   post_title      12730 non-null  object
1   post_url        12730 non-null  object
2   comment         12503 non-null  object
3   post_body       227 non-null    object
4   created_utc     12730 non-null  object
5   post_full       12730 non-null  object
dtypes: object(6)
memory usage: 596.8+ KB
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 121 entries, 0 to 120
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   title            121 non-null    object
1   link             121 non-null    object
2   date_published   121 non-null    object
3   text             121 non-null    object
4   cleaned_text     121 non-null    object
dtypes: object(5)
memory usage: 4.9+ KB
```

```
[ ]: # Stopwords

import nltk
from nltk.corpus import stopwords

nltk.download("stopwords")
english_stop = stopwords.words("english")

filipino_stopwords = [
    "sa", "ang", "ng", "mga", "si", "ni", "kay", "kina",
    "ito", "iyan", "iyon", "doon", "diyan", "dito",
    "ko", "mo", "nya", "niya", "natin", "namin", "nyo", "ninyo",
    "sila", "kami", "tayo", "kayo",
    "ay", "may", "wala", "meron", "naman", "rin", "din",
    "pa", "ba", "po", "ho", "eh", "yung", "yup", "nope",
    "kung", "kapag", "pero", "kasi", "para", "habang", "bago",
    "nga", "lang", "daw", "raw", "yata", "pala", "talaga",
    "siguro", "naka", "ni", "siya",
```

```

# noise words
"haha", "hahaha", "hehe", "lol", "omg", "hmmmm", "hmmmmmm",
"tsk", "hay", "amp", "grabe",

# more noise as the code is ran
"lalala", "brgy", "sen", "ph", "phl", "gov",
"committee", "chairmanships", "said", "s",
]

combined_stopwords = list(set(english_stop + filipino_stopwords))

```

```

[nltk_data] Downloading package stopwords to
[nltk_data] C:\Users\Uwie\AppData\Roaming\nltk_data...
[nltk_data] Package stopwords is already up-to-date!

```

Word Cloud based on Word Frequency

```

[ ]: def generate_wordcloud(text_series, title, extra_stopwords=None):
    all_text = ' '.join(text_series)

    stopwords = set(STOPWORDS)
    if extra_stopwords:
        stopwords.update(extra_stopwords).union(combined_stopwords)

    wordcloud = WordCloud(width=800, height=400, background_color='white',
        stopwords=stopwords).generate(all_text)

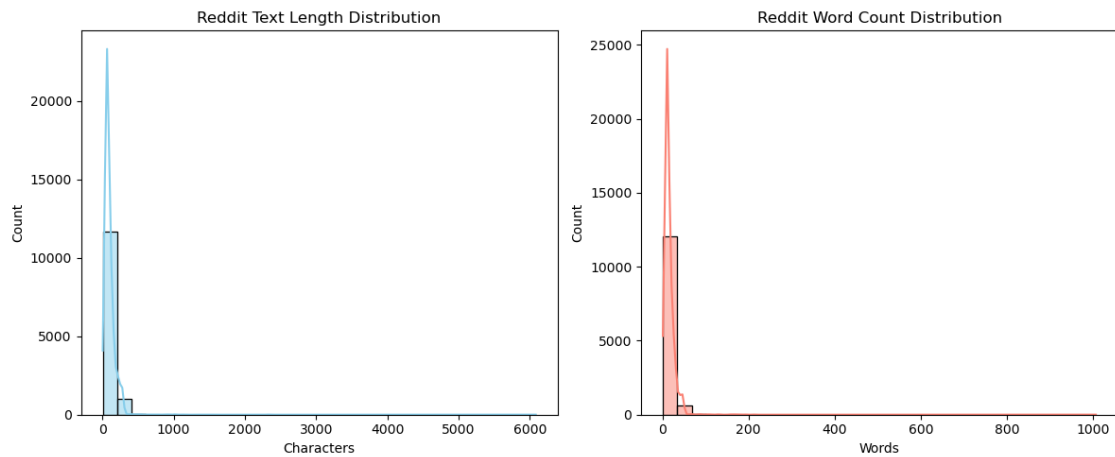
    plt.figure(figsize=(10,5))
    plt.imshow(wordcloud, interpolation='bilinear')
    plt.axis('off')
    plt.title(title, fontsize=16)
    plt.show()

# --- Function to compute text stats ---
def add_text_stats(df, column):
    df['text_length'] = df[column].apply(len)
    df['word_count'] = df[column].apply(lambda x: len(x.split()))
    return df

# --- Function to plot Seaborn visualizations ---
def plot_text_stats(df, dataset_name):
    plt.figure(figsize=(12,5))

    # Text length histogram
    plt.subplot(1,2,1)
    sns.histplot(df['text_length'], bins=30, kde=True, color='skyblue')

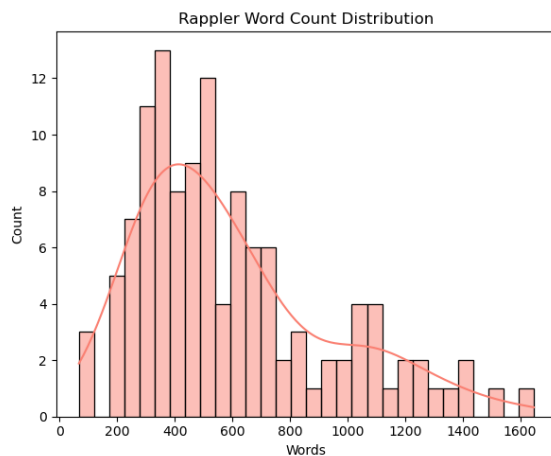
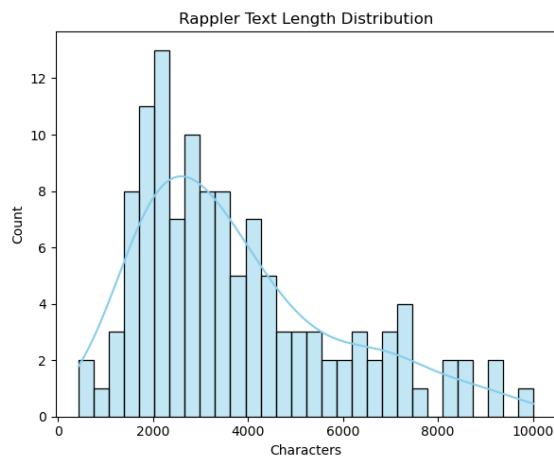
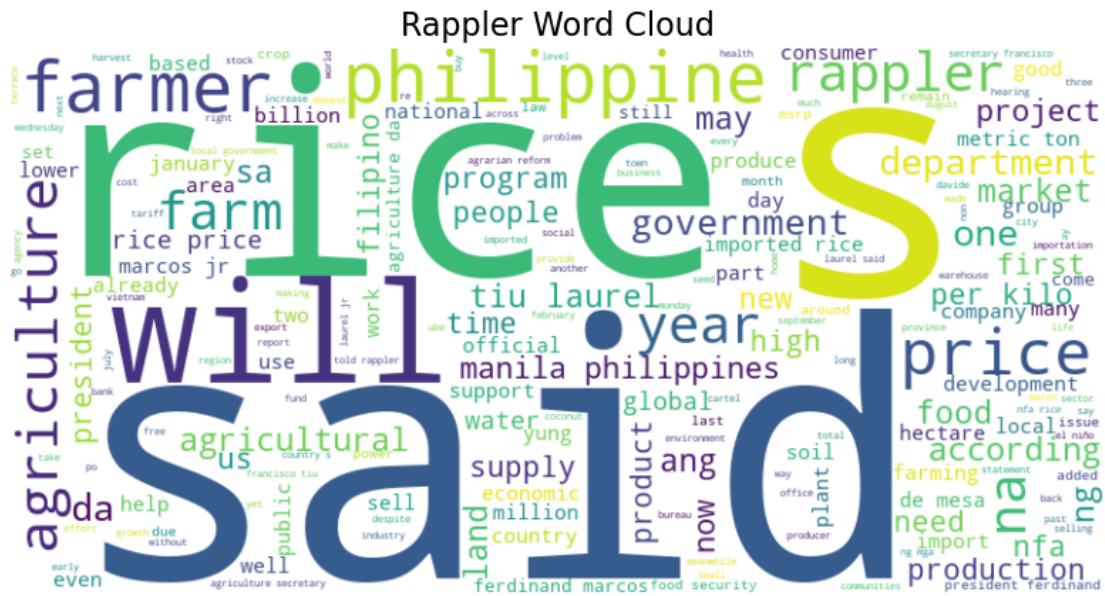
```

```
[ ]: rappler_clean = add_text_stats(rappler_clean, "cleaned_text")
print("Rappler Text Stats Summary:")
print(rappler_clean[['text_length', 'word_count']].describe())
generate_wordcloud(rappler_clean['cleaned_text'], "Rappler Word Cloud")
plot_text_stats(rappler_clean, "Rappler")
```

Rappler Text Stats Summary:

	text_length	word_count
count	121.000000	121.000000
mean	3791.082645	595.752066
std	2110.447329	334.902630
min	437.000000	67.000000
25%	2191.000000	341.000000
50%	3295.000000	509.000000
75%	4873.000000	738.000000
max	10015.000000	1647.000000



Getting top words

```
[ ]: from collections import Counter
import nltk
from nltk.corpus import stopwords
nltk.download('stopwords')

def get_top_words(text_series, n=20, extra_stopwords=None):
    """
    Returns top n frequent words (excluding stopwords)
```

```

"""
all_text = ' '.join(text_series).lower().split()
stop_words_set = set(stopwords.words('english'))
if extra_stopwords:
    stop_words_set.update(extra_stopwords)

filtered_words = [w for w in all_text if w.isalpha() and w not in
↳stop_words_set]
word_freq = Counter(filtered_words)
return word_freq.most_common(n)

```

```

[nltk_data] Downloading package stopwords to
[nltk_data] C:\Users\Uwie\AppData\Roaming\nltk_data...
[nltk_data] Package stopwords is already up-to-date!

```

```

[ ]: reddit_top_words = get_top_words(reddit_clean['post_full'], n=20)
print("Reddit Top 20 Words:")
print(reddit_top_words)

rappler_top_words = get_top_words(rappler_clean['cleaned_text'], n=20)
print("Rappler Top 20 Words:")
print(rappler_top_words)

```

Reddit Top 20 Words:

```

[('agriculture', 2568), ('sa', 2002), ('na', 1554), ('ang', 1302), ('ng', 1264),
('rice', 1118), ('agricultural', 1112), ('philippines', 1005), ('profession',
918), ('pa', 829), ('ba', 782), ('department', 636), ('may', 634), ('cynthia',
632), ('committee', 601), ('senate', 597), ('mga', 593), ('one', 591),
('marcos', 581), ('farmers', 565)]

```

Rappler Top 20 Words:

```

[('rice', 478), ('said', 405), ('ng', 223), ('agriculture', 215), ('food', 205),
('farmers', 186), ('philippines', 183), ('sa', 182), ('also', 181), ('na', 163),
('prices', 142), ('government', 138), ('agricultural', 138), ('price', 138),
('department', 132), ('ang', 129), ('would', 123), ('da', 120), ('marcos', 117),
('per', 117)]

```

```

[ ]: # Define Filipino stopwords
filipino_stopwords = {'na', 'sa', 'ang', 'mga', 'pa', 'ba', 'ng', 'ay', 'si',
↳'ni', 'kay', 'may', 'also', 'mas'}

# Get top words excluding English + Filipino stopwords
reddit_top_words = get_top_words(
    reddit_clean['post_full'],
    n=20,
    extra_stopwords=filipino_stopwords
)
print("Reddit Top 20 Words (Filipino Stopwords Removed):")
print(reddit_top_words)

```

```

rappler_top_words = get_top_words(
    rappler_clean['cleaned_text'],
    n=20,
    extra_stopwords=filipino_stopwords
)
print("Rappler Top 20 Words (Filipino Stopwords Removed):")
print(rappler_top_words)

```

Reddit Top 20 Words (Filipino Stopwords Removed):

```

[('agriculture', 2568), ('rice', 1118), ('agricultural', 1112), ('philippines',
1005), ('profession', 918), ('department', 636), ('cynthia', 632), ('committee',
601), ('senate', 597), ('one', 591), ('marcos', 581), ('farmers', 565), ('goes',
563), ('clearance', 560), ('fish', 505), ('least', 496), ('bakit', 495),
('live', 480), ('list', 479), ('field', 479)]

```

Rappler Top 20 Words (Filipino Stopwords Removed):

```

[('rice', 478), ('said', 405), ('agriculture', 215), ('food', 205), ('farmers',
186), ('philippines', 183), ('prices', 142), ('government', 138),
('agricultural', 138), ('price', 138), ('department', 132), ('would', 123),
('da', 120), ('marcos', 117), ('per', 117), ('laurel', 116), ('local', 108),
('one', 102), ('according', 100), ('president', 100)]

```

Post frequency over time

```

[ ]: import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt

def plot_posts_over_time(df, date_column, dataset_name):
    """
    Plots number of posts/articles per day using Seaborn
    """
    # Ensure datetime
    df[date_column] = pd.to_datetime(df[date_column], errors='coerce')

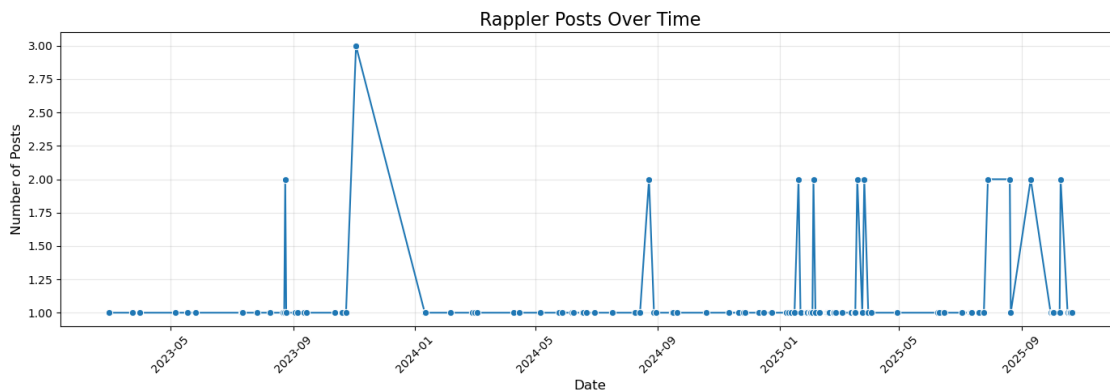
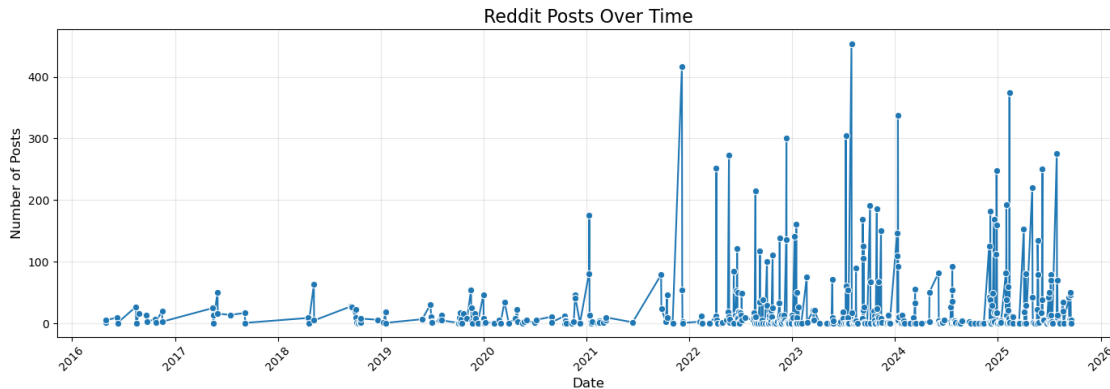
    # Count posts per day
    ts = df.groupby(df[date_column].dt.date).size().reset_index(name='count')
    ts.rename(columns={date_column: 'date'}, inplace=True)

    # Plot with Seaborn
    plt.figure(figsize=(14,5))
    sns.lineplot(data=ts, x='date', y='count', marker='o')
    plt.title(f'{dataset_name} Posts Over Time', fontsize=16)
    plt.xlabel('Date', fontsize=12)
    plt.ylabel('Number of Posts', fontsize=12)
    plt.xticks(rotation=45)

```

```
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

```
[ ]: plot_posts_over_time(reddit_clean, 'created_utc', 'Reddit')
plot_posts_over_time(rappler_clean, 'date_published', 'Rappler')
```



Summary of EDA

The EDA included: 1. Inspecting dataset structure using `data.info()`. 2. Generating a basic (but still noisy) word cloud from raw word frequencies. 3. Identifying the top most frequent words. 4. Plotting post frequency over time to observe activity trends.

6.1 Word Embeddings

```
[ ]: !pip install scikit-learn pandas numpy
```

Requirement already satisfied: scikit-learn in
 /Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-

packages (1.7.2)
Requirement already satisfied: pandas in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (2.3.3)
Requirement already satisfied: numpy in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (2.3.3)
Requirement already satisfied: scipy>=1.8.0 in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (from scikit-learn) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (from scikit-learn) (3.6.0)
Requirement already satisfied: python-dateutil>=2.8.2 in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in
/Users/uwie/Documents/_Github/CSCI161-SocialComputing/.venv/lib/python3.13/site-
packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

[notice] A new release of pip is
available: 25.2 -> 25.3
[notice] To update, run:
pip install --upgrade pip

```
[ ]: import pandas as pd
import numpy as np
from sklearn.feature_extraction.text import CountVectorizer
import matplotlib.pyplot as plt

# Load Reddit
reddit_corpus = pd.read_csv("../data/Reddit/reddit_clean.csv")
print(reddit_corpus.tail())

# Load Rappler
rappler_corpus = pd.read_csv("../data/News/rappler_clean.csv")
print(rappler_corpus.tail())
```

post_title \

12725 Philippines, Japan deepen ties with \$1.5 billi...
 12726 Philippines, Japan deepen ties with \$1.5 billi...
 12727 Philippines, Japan deepen ties with \$1.5 billi...
 12728 Philippines, Japan deepen ties with \$1.5 billi...
 12729 Philippines, Japan deepen ties with \$1.5 billi...

post_url \
 12725 https://reddit.com/r/Philippines/comments/1cio...
 12726 https://reddit.com/r/Philippines/comments/1cio...
 12727 https://reddit.com/r/Philippines/comments/1cio...
 12728 https://reddit.com/r/Philippines/comments/1cio...
 12729 https://reddit.com/r/Philippines/comments/1cio...

comment post_body \
 12725 that's why fuck the dutertes NaN
 12726 fr? that's scary. NaN
 12727 this partnership is very daijoubu NaN
 12728 yeah, there's a huge connection between corrup... NaN
 12729 yup ganyan din ginagawa ng russia, kaya mas pi... NaN

created_utc post_full
 12725 2024-05-03 11:30:11 philippines, japan deepen ties with \$1.5 billi...
 12726 2024-05-03 12:32:41 philippines, japan deepen ties with \$1.5 billi...
 12727 2024-05-03 10:33:52 philippines, japan deepen ties with \$1.5 billi...
 12728 2024-05-03 12:36:12 philippines, japan deepen ties with \$1.5 billi...
 12729 2024-05-03 16:57:35 philippines, japan deepen ties with \$1.5 billi...

title \
 116 ASF spreads to 4 Aklan towns
 117 Pangasinan farmers urged to plant early as San...
 118 [ANALYSIS] Should LBP and DBP be merged into a...
 119 Sugar regulation chief Alba resigns
 120 North Korea's Kim orders 'fundamental transfor...

link \
 116 https://www.rappler.com/philippines/visayas/as...
 117 https://www.rappler.com/philippines/luzon/pang...
 118 https://www.rappler.com/voices/thought-leaders...
 119 https://www.rappler.com/business/sugar-regulat...
 120 https://www.rappler.com/world/asia-pacific/nor...

date_published \
 116 2023-05-18T09:10:00+08:00
 117 2023-05-06T11:32:23+08:00
 118 2023-03-31T13:24:41+08:00
 119 2023-03-24T11:19:13+08:00
 120 2023-02-28T11:37:49+08:00

text \
 116


```

116 AKLAN, Philippines - A month after the first c...
117 DAGUPAN CITY, Philippines - With El Niño just ...
118 On March 23, Finance Secretary Benjamin Diokno...
119 MANILA, Philippines - Sugar Regulatory Adminis...
120 SEOUL, South Korea - North Korean leaderKim Jo...

```

```

                                cleaned_text
116 aklan, philippines - a month after the first c...
117 dagupan city, philippines - with el niño just ...
118 on march 23, finance secretary benjamin diokno...
119 manila, philippines - sugar regulatory adminis...
120 seoul, south korea - north korean leaderkim jo...

```

```

[ ]: reddit_texts = reddit_corpus["post_full"].astype(str).tolist()
     rappler_texts = rappler_corpus["text"].astype(str).tolist()

```

```

[ ]: # Stopwrods

import nltk
from nltk.corpus import stopwords

nltk.download("stopwords")
english_stop = stopwords.words("english")

filipino_stopwords = [
    "sa", "ang", "ng", "mga", "si", "ni", "kay", "kina",
    "ito", "iyan", "iyon", "doon", "diyan", "dito",
    "ko", "mo", "nya", "niya", "natin", "namin", "nyo", "ninyo",
    "sila", "kami", "tayo", "kayo",
    "ay", "may", "wala", "meron", "naman", "rin", "din",
    "pa", "ba", "po", "ho", "eh", "yung", "yup", "nope",
    "kung", "kapag", "pero", "kasi", "para", "habang", "bago",
    "nga", "lang", "daw", "raw", "yata", "pala", "talaga",
    "siguro", "naka", "ni", "siya",

    # noise words
    "haha", "hahaha", "hehe", "lol", "omg", "hmmmm", "hmmmmmm",
    "tsk", "hay", "amp", "grabe",

    # more noise as the code is ran
    "lalala", "brgy", "sen", "ph", "phl", "gov",
    "committee", "chairmanships"
]

combined_stopwords = list(set(english_stop + filipino_stopwords))

```

[nltk_data] Downloading package stopwords to /Users/uwie/nltk_data...

[nltk_data] Package stopwords is already up-to-date!

Reddit BoW

```
[ ]: from sklearn.feature_extraction.text import CountVectorizer

# Use only the combined post text (title + body)
texts_reddit = reddit_corpus["post_full"].astype(str)

vectorizer = CountVectorizer(
    stop_words=combined_stopwords,
    max_df=0.95,
    min_df=2,
    token_pattern=r'(?u)\b[a-zA-Z][a-zA-Z]+\b'  # remove numbers, symbols,
    ↪ 1-letter tokens
)

reddit_bow = vectorizer.fit_transform(texts_reddit)
reddit_bow.shape
```

```
[ ]: (12730, 1790)
```

```
[ ]: from wordcloud import WordCloud
import numpy as np
import matplotlib.pyplot as plt

# Get vocabulary + term frequencies
vocab = vectorizer.get_feature_names_out()
word_freq = np.array(reddit_bow.sum(axis=0)).flatten()

freq_dict = dict(zip(vocab, word_freq))

# Generate the cleaned, BoW-based wordcloud
plt.figure(figsize=(15, 7))
wc = WordCloud(
    width=1200,
    height=600,
    background_color="white"
)
plt.imshow(wc.generate_from_frequencies(freq_dict))
plt.axis("off")
plt.show()
```



```

freq_dict = dict(zip(vocab, word_freq))

# Generate the cleaned, BoW-based wordcloud
plt.figure(figsize=(15, 7))
wc = WordCloud(
    width=1200,
    height=600,
    background_color="white"
)
plt.imshow(wc.generate_from_frequencies(freq_dict))
plt.axis("off")
plt.show()

```



LDA, Topic Modelling for Reddit

```

[ ]: from sklearn.decomposition import LatentDirichletAllocation

reddit_lda_model = LatentDirichletAllocation(
    n_components=5,
    random_state=42,
    learning_method='batch'
)
reddit_lda_model.fit(reddit_bow)

[ ]: LatentDirichletAllocation(n_components=5, random_state=42)

```

```
[ ]: def show_topics(model, feature_names, n_top_words=12):
    for idx, topic in enumerate(model.components_):
        top_indices = topic.argsort()[::-n_top_words - 1:-1]
        top_words = [feature_names[i] for i in top_indices]
        print(f"\nTopic {idx+1}:")
        print(" | ".join(top_words))

feature_names = vectorizer.get_feature_names_out()
show_topics(reddit_lda_model, feature_names)
```

Topic 1:

lot | africa | integrated | cases | importing | jayson | fever | announced |
million | representative | affordable | de

Topic 2:

lot | africa | done | hectare | science | businesses | living | different |
integrated | importations | financial | immediately

Topic 3:

heat | laurel | growing | lot | distributed | irrigation | million |
representative | parents | energy | industriya | integrated

Topic 4:

lot | affordable | october | records | landed | millions | harvest | group | di
| children | among | accessible

Topic 5:

africa | operates | determined | personal | association | play | quimbo |
establish | promote | mesa | result | customs

```
[ ]: topic_distribution = reddit_lda_model.transform(reddit_bow)

reddit_corpus["dominant_topic"] = topic_distribution.argmax(axis=1)
reddit_corpus["topic_score"] = topic_distribution.max(axis=1)

reddit_corpus.to_csv("../data/Reddit/reddit_with_topics.csv", index=False)
```

LDA, Topic Modelling for Rappler

```
[ ]: from sklearn.decomposition import LatentDirichletAllocation

rappler_lda_model = LatentDirichletAllocation(
    n_components=5,
    random_state=42,
    learning_method='batch'
)
```

```
rappler_lda_model.fit(rappler_bow)
```

```
[ ]: LatentDirichletAllocation(n_components=5, random_state=42)
```

```
[ ]: def show_topics(model, feature_names, n_top_words=12):  
    for idx, topic in enumerate(model.components_):  
        top_indices = topic.argsort()[: -n_top_words - 1:-1]  
        top_words = [feature_names[i] for i in top_indices]  
        print(f"\nTopic {idx+1}:")  
        print(" | ".join(top_words))
```

```
feature_names = vectorizer.get_feature_names_out()  
show_topics(rappler_lda_model, feature_names)
```

Topic 1:

farmers | farm | food | one | rice | like | also | farming | years | soil |
people | agricultural

Topic 2:

price | rice | prices | also | marcos | president | agriculture | year | supply
| august | government | economic

Topic 3:

na | agriculture | laurel | marcos | farmers | land | agricultural | sector |
reform | president | agrarian | also

Topic 4:

rice | da | prices | nfa | kilo | food | per | imported | farmers | agriculture
| price | government

Topic 5:

agriculture | government | would | also | project | according | farmers |
irrigation | department | production | philippine | national

```
[ ]: topic_distribution = rappler_lda_model.transform(rappler_bow)  
  
rappler_corpus["dominant_topic"] = topic_distribution.argmax(axis=1)  
rappler_corpus["topic_score"] = topic_distribution.max(axis=1)  
  
rappler_corpus.to_csv("../data/News/rappler_with_topics.csv", index=False)
```

Checkpoint Data Viewing

```
[ ]: rappler_withTopic = pd.read_csv("../data/News/rappler_with_topics.csv")
      reddit_withTopic = pd.read_csv("../data/Reddit/reddit_with_topics.csv")

      rappler_withTopic.tail()
```

```
[ ]:
116                                     title \
117                                     ASF spreads to 4 Aklan towns
118 Pangasinan farmers urged to plant early as San...
118 [ANALYSIS] Should LBP and DBP be merged into a...
119                                     Sugar regulation chief Alba resigns
120 North Korea's Kim orders 'fundamental transfor...

                                     link \
116 https://www.rappler.com/philippines/visayas/as...
117 https://www.rappler.com/philippines/luzon/pang...
118 https://www.rappler.com/voices/thought-leaders...
119 https://www.rappler.com/business/sugar-regulat...
120 https://www.rappler.com/world/asia-pacific/nor...

                                     date_published \
116 2023-05-18T09:10:00+08:00
117 2023-05-06T11:32:23+08:00
118 2023-03-31T13:24:41+08:00
119 2023-03-24T11:19:13+08:00
120 2023-02-28T11:37:49+08:00

                                     text \
116 AKLAN, Philippines - A month after the first c...
117 DAGUPAN CITY, Philippines - With El Niño just ...
118 On March 23, Finance Secretary Benjamin Diokno...
119 MANILA, Philippines - Sugar Regulatory Adminis...
120 SEOUL, South Korea - North Korean leaderKim Jo...

                                     cleaned_text  dominant_topic \
116 aklan, philippines - a month after the first c...          1
117 dagupan city, philippines - with el niño just ...          4
118 on march 23, finance secretary benjamin diokno...          2
119 manila, philippines - sugar regulatory adminis...          1
120 seoul, south korea - north korean leaderkim jo...          4

                                     topic_score
116      0.992306
117      0.653853
118      0.436670
119      0.994174
120      0.658819
```

```
[ ]: reddit_withTopic.tail()
```

```
[ ]:                                     post_title \
12725  Philippines, Japan deepen ties with $1.5 billi...
12726  Philippines, Japan deepen ties with $1.5 billi...
12727  Philippines, Japan deepen ties with $1.5 billi...
12728  Philippines, Japan deepen ties with $1.5 billi...
12729  Philippines, Japan deepen ties with $1.5 billi...

                                     post_url \
12725  https://reddit.com/r/Philippines/comments/1cio...
12726  https://reddit.com/r/Philippines/comments/1cio...
12727  https://reddit.com/r/Philippines/comments/1cio...
12728  https://reddit.com/r/Philippines/comments/1cio...
12729  https://reddit.com/r/Philippines/comments/1cio...

                                     comment post_body \
12725                                     that's why fuck the dutertes      NaN
12726                                     fr? that's scary.                  NaN
12727                                     this partnership is very daijoubu    NaN
12728  yeah, there's a huge connection between corrup...      NaN
12729  yup ganyan din ginagawa ng russia, kaya mas pi...      NaN

                                     created_utc                                     post_full \
12725  2024-05-03 11:30:11  philippines, japan deepen ties with $1.5 billi...
12726  2024-05-03 12:32:41  philippines, japan deepen ties with $1.5 billi...
12727  2024-05-03 10:33:52  philippines, japan deepen ties with $1.5 billi...
12728  2024-05-03 12:36:12  philippines, japan deepen ties with $1.5 billi...
12729  2024-05-03 16:57:35  philippines, japan deepen ties with $1.5 billi...

                                     dominant_topic  topic_score
12725                                     3          0.899632
12726                                     3          0.899632
12727                                     3          0.899632
12728                                     3          0.899632
12729                                     3          0.899632
```

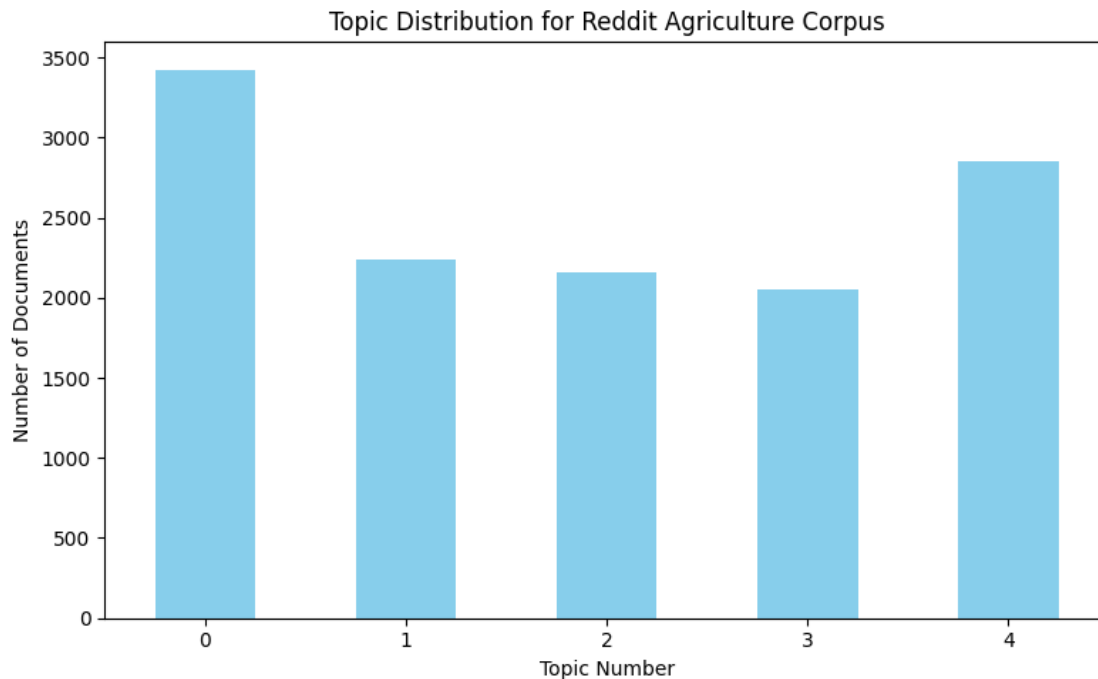
```
[ ]: import matplotlib.pyplot as plt

# Count how many posts fall under each topic
reddit_topic_counts = reddit_corpus["dominant_topic"].value_counts().
    ↪sort_index()

plt.figure(figsize=(8,5))
reddit_topic_counts.plot(kind="bar", color="skyblue")
plt.xlabel("Topic Number")
```

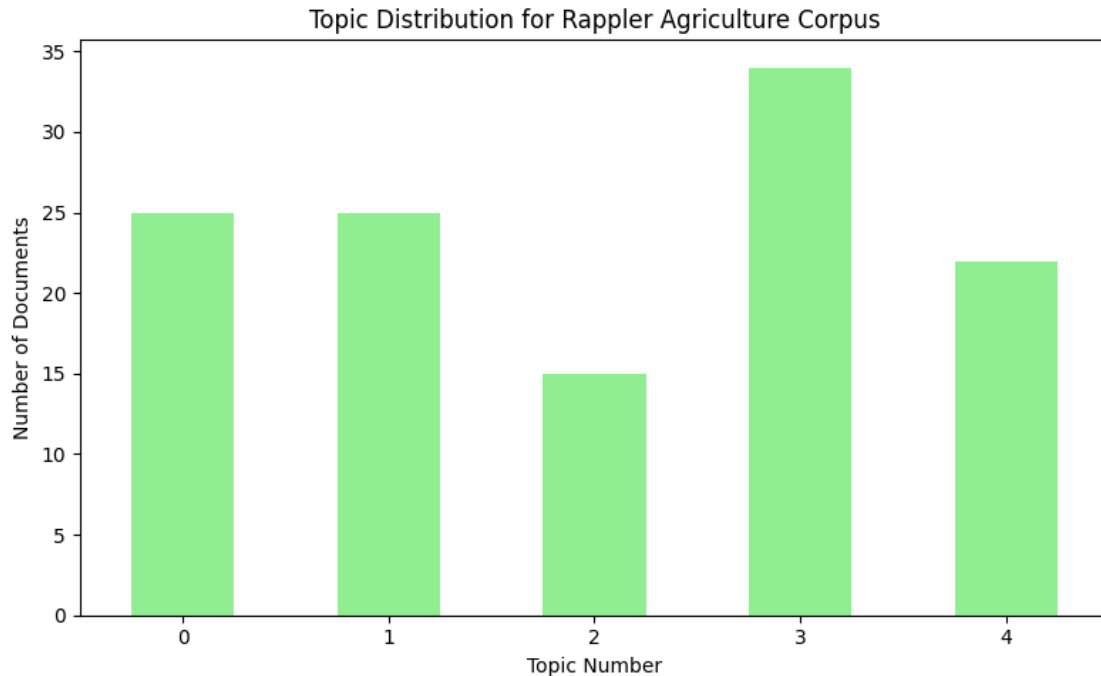


```
plt.ylabel("Number of Documents")
plt.title("Topic Distribution for Reddit Agriculture Corpus")
plt.xticks(rotation=0)
plt.tight_layout()
plt.savefig("../IEEE/figures/reddit_topic_distribution.png", dpi=300)
plt.show()
```



```
[ ]: rappler_topic_counts = rappler_corpus["dominant_topic"].value_counts().
    ↪sort_index()

plt.figure(figsize=(8,5))
rappler_topic_counts.plot(kind="bar", color="lightgreen")
plt.xlabel("Topic Number")
plt.ylabel("Number of Documents")
plt.title("Topic Distribution for Rappler Agriculture Corpus")
plt.xticks(rotation=0)
plt.tight_layout()
plt.savefig("../IEEE/figures/rappler_topic_distribution.png", dpi=300)
plt.show()
```



6.2 Sentiment Analysis

I want to approach this through BERT. The `multilingual-sentiment-analysis` model by `tabularisai` seems to be best fit as my dataset contains tagalog and english contents. Although it shows more english-comments/posts, it is necessary to try to take into account the tagalog content.

Setup

```
[ ]: !pip install transformers
```

```
Requirement already satisfied: transformers in c:\users\uwie\anaconda3\lib\site-packages (4.44.2)
Requirement already satisfied: filelock in c:\users\uwie\anaconda3\lib\site-packages (from transformers) (3.13.1)
Requirement already satisfied: huggingface-hub<1.0,>=0.23.2 in c:\users\uwie\anaconda3\lib\site-packages (from transformers) (0.35.0)
Requirement already satisfied: numpy>=1.17 in c:\users\uwie\anaconda3\lib\site-packages (from transformers) (1.26.4)
Requirement already satisfied: packaging>=20.0 in c:\users\uwie\anaconda3\lib\site-packages (from transformers) (24.2)
Requirement already satisfied: pyyaml>=5.1 in c:\users\uwie\anaconda3\lib\site-packages (from transformers) (6.0.2)
Requirement already satisfied: regex!=2019.12.17 in c:\users\uwie\anaconda3\lib\site-packages (from transformers) (2024.11.6)
```

Requirement already satisfied: requests in c:\users\uwie\anaconda3\lib\site-packages (from transformers) (2.32.3)
 Requirement already satisfied: safetensors>=0.4.1 in c:\users\uwie\anaconda3\lib\site-packages (from transformers) (0.6.2)
 Requirement already satisfied: tokenizers<0.20,>=0.19 in c:\users\uwie\anaconda3\lib\site-packages (from transformers) (0.19.1)
 Requirement already satisfied: tqdm>=4.27 in c:\users\uwie\anaconda3\lib\site-packages (from transformers) (4.67.1)
 Requirement already satisfied: fsspec>=2023.5.0 in c:\users\uwie\anaconda3\lib\site-packages (from huggingface-hub<1.0,>=0.23.2->transformers) (2024.12.0)
 Requirement already satisfied: typing-extensions>=3.7.4.3 in c:\users\uwie\anaconda3\lib\site-packages (from huggingface-hub<1.0,>=0.23.2->transformers) (4.12.2)
 Requirement already satisfied: colorama in c:\users\uwie\anaconda3\lib\site-packages (from tqdm>=4.27->transformers) (0.4.6)
 Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\uwie\anaconda3\lib\site-packages (from requests->transformers) (3.3.2)
 Requirement already satisfied: idna<4,>=2.5 in c:\users\uwie\anaconda3\lib\site-packages (from requests->transformers) (3.7)
 Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\uwie\anaconda3\lib\site-packages (from requests->transformers) (2.3.0)
 Requirement already satisfied: certifi>=2017.4.17 in c:\users\uwie\anaconda3\lib\site-packages (from requests->transformers) (2025.1.31)

[notice] A new release of pip is available: 25.2 -> 25.3

[notice] To update, run: python.exe -m pip install --upgrade pip

```
[ ]: import torch
      print(torch.cuda.is_available())
      print(torch.cuda.get_device_name(0))
```

True

NVIDIA GeForce RTX 3070

```
[ ]: from transformers import AutoTokenizer, AutoModelForSequenceClassification
      import torch

      tokenizer = AutoTokenizer.from_pretrained("tabularisai/
      ↪multilingual-sentiment-analysis")
      model = AutoModelForSequenceClassification.from_pretrained("tabularisai/
      ↪multilingual-sentiment-analysis")

      device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
      model = model.to(device)
      model.eval()
```

```
[ ]: DistilBertForSequenceClassification(
  (distilbert): DistilBertModel(
    (embeddings): Embeddings(
      (word_embeddings): Embedding(119547, 768, padding_idx=0)
      (position_embeddings): Embedding(512, 768)
      (LayerNorm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
      (dropout): Dropout(p=0.1, inplace=False)
    )
    (transformer): Transformer(
      (layer): ModuleList(
        (0-5): 6 x TransformerBlock(
          (attention): MultiHeadSelfAttention(
            (dropout): Dropout(p=0.1, inplace=False)
            (q_lin): Linear(in_features=768, out_features=768, bias=True)
            (k_lin): Linear(in_features=768, out_features=768, bias=True)
            (v_lin): Linear(in_features=768, out_features=768, bias=True)
            (out_lin): Linear(in_features=768, out_features=768, bias=True)
          )
          (sa_layer_norm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
          (ffn): FFN(
            (dropout): Dropout(p=0.1, inplace=False)
            (lin1): Linear(in_features=768, out_features=3072, bias=True)
            (lin2): Linear(in_features=3072, out_features=768, bias=True)
            (activation): GELUActivation()
          )
          (output_layer_norm): LayerNorm((768,), eps=1e-12,
elementwise_affine=True)
        )
      )
    )
  )
  (pre_classifier): Linear(in_features=768, out_features=768, bias=True)
  (classifier): Linear(in_features=768, out_features=5, bias=True)
  (dropout): Dropout(p=0.2, inplace=False)
)
```

Run the code

```
[ ]: import numpy as np
import pandas as pd

id2label = model.config.id2label

def get_sentiment(text):
    if not isinstance(text, str):
        text = str(text)
```

```

# Tokenize and move input tensors to GPU
inputs = tokenizer(
    text,
    truncation=True,
    max_length=512,
    return_tensors="pt"
).to(device)

with torch.no_grad():
    outputs = model(**inputs)
    scores = outputs.logits[0]
    label_id = int(torch.argmax(scores))
    confidence = float(torch.softmax(scores, dim=0)[label_id])

return id2label[label_id], confidence

```

```

[ ]: rappler_corpus = pd.read_csv("../data/News/rappler_with_topics.csv")
     reddit_corpus = pd.read_csv("../data/Reddit/reddit_with_topics.csv")

```

```

[ ]: reddit_corpus["sentiment_label"], reddit_corpus["sentiment_score"] = zip(
     *reddit_corpus["post_full"].astype(str).apply(get_sentiment)
)

```

```

[ ]: rappler_corpus["sentiment_label"], rappler_corpus["sentiment_score"] = zip(
     *rappler_corpus["cleaned_text"].astype(str).apply(get_sentiment)
)

```

Sentiment Results

```

[ ]: reddit_corpus["sentiment_label"].value_counts()

```

```

[ ]: sentiment_label
Neutral      8368
Negative     2802
Very Negative 1098
Positive      350
Very Positive  112
Name: count, dtype: int64

```

```

[ ]: rappler_corpus["sentiment_label"].value_counts()

```

```

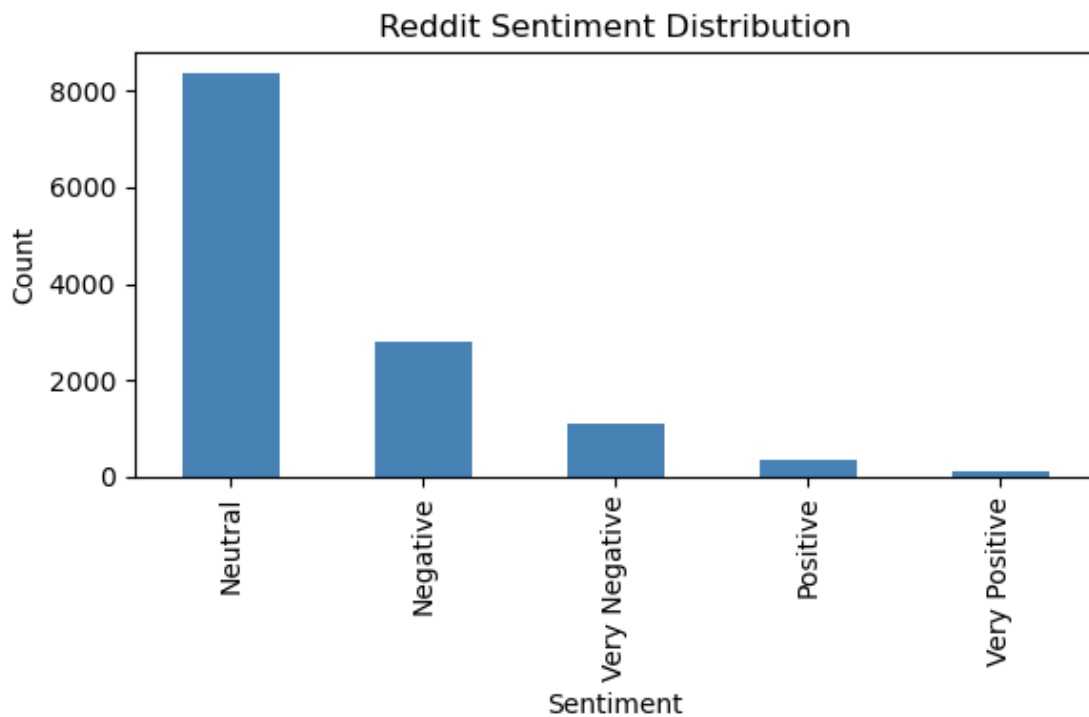
[ ]: sentiment_label
Neutral      113
Negative       6
Very Positive  2

```

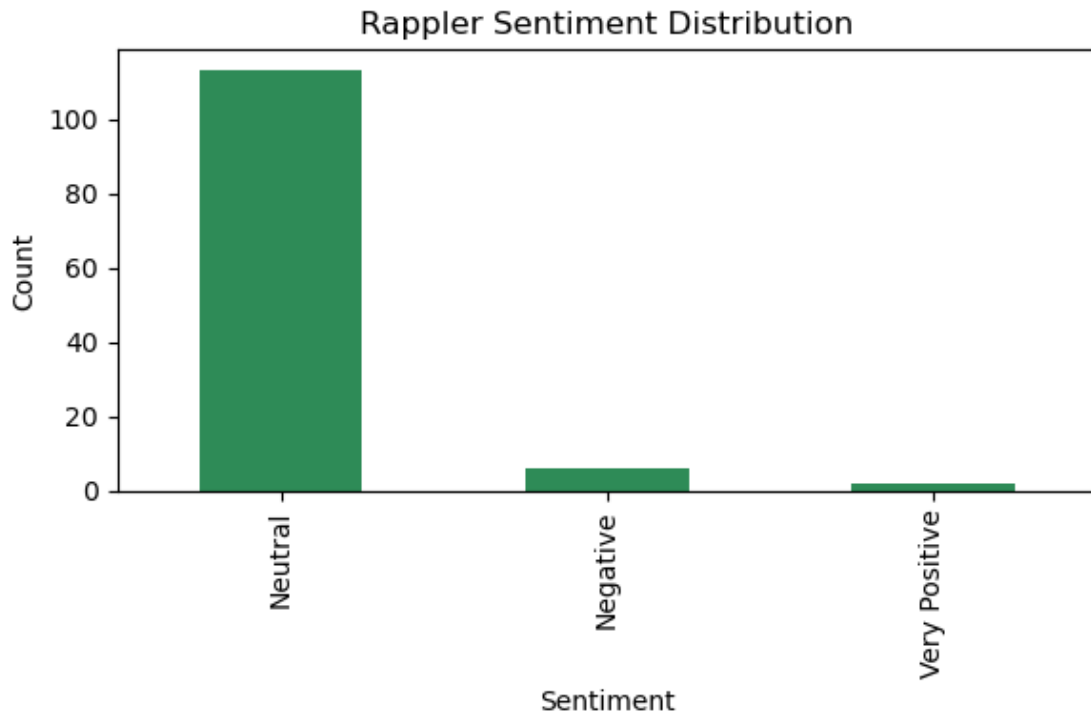
Name: count, dtype: int64

```
[ ]: import matplotlib.pyplot as plt

plt.figure(figsize=(6,4))
reddit_corpus["sentiment_label"].value_counts().plot(kind="bar",
    color="steelblue")
plt.title("Reddit Sentiment Distribution")
plt.xlabel("Sentiment")
plt.ylabel("Count")
plt.tight_layout()
plt.savefig("../IEEE/figures/reddit_sentiment_distribution.png", dpi=300)
plt.show()
```



```
[ ]: plt.figure(figsize=(6,4))
rappler_corpus["sentiment_label"].value_counts().plot(kind="bar",
    color="seagreen")
plt.title("Rappler Sentiment Distribution")
plt.xlabel("Sentiment")
plt.ylabel("Count")
plt.tight_layout()
plt.savefig("../IEEE/figures/rappler_sentiment_distribution.png", dpi=300)
plt.show()
```



Topic and Sentiment Overlaps

```
[ ]: reddit_cross = pd.crosstab(
    reddit_corpus["dominant_topic"],
    reddit_corpus["sentiment_label"],
    normalize="index"    # percent per topic
)

reddit_cross
```

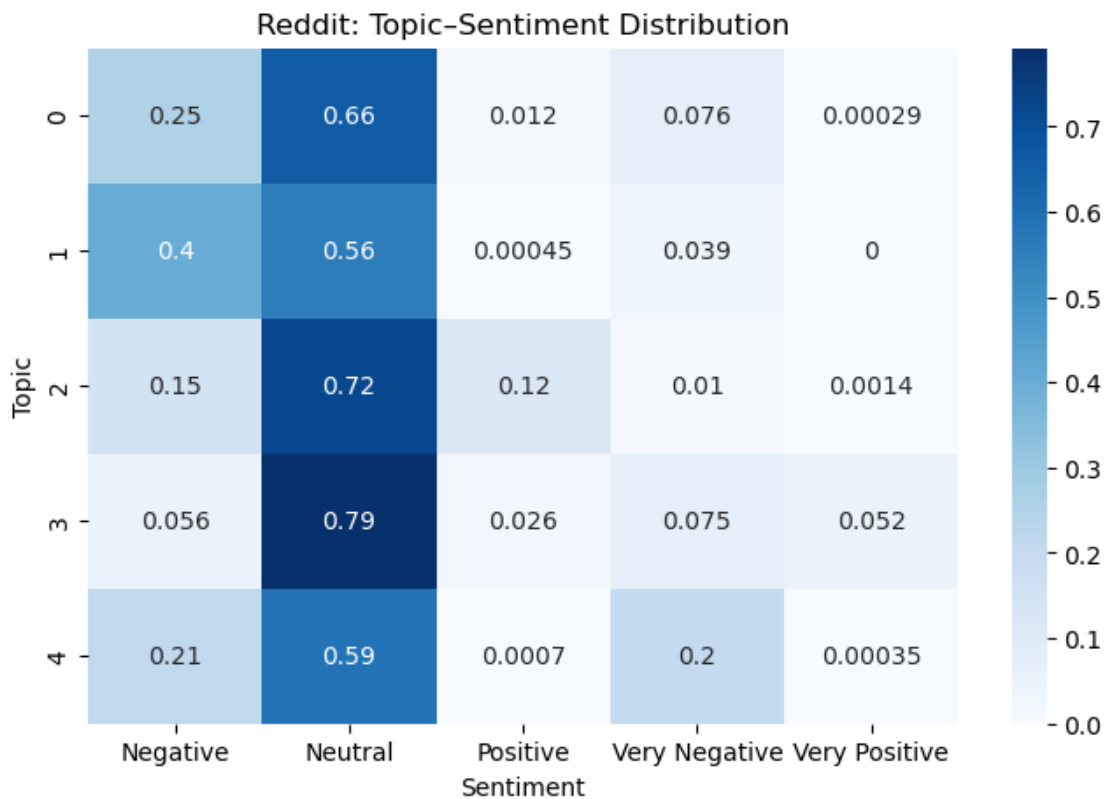
```
[ ]: sentiment_label  Negative    Neutral    Positive  Very Negative  Very Positive
dominant_topic
0          0.251095  0.660730  0.011971          0.075912          0.000292
1          0.402586  0.557735  0.000446          0.039233          0.000000
2          0.150463  0.720833  0.117130          0.010185          0.001389
3          0.055556  0.791910  0.025828          0.074561          0.052144
4          0.210526  0.586667  0.000702          0.201754          0.000351
```

```
[ ]: rappler_cross = pd.crosstab(
    rappler_corpus["dominant_topic"],
    rappler_corpus["sentiment_label"],
    normalize="index"
```

```
)
rappler_cross
```

```
[ ]: sentiment_label  Negative    Neutral    Very Positive
dominant_topic
0                0.000000    0.960000         0.040000
1                0.040000    0.960000         0.000000
2                0.066667    0.866667         0.066667
3                0.029412    0.970588         0.000000
4                0.136364    0.863636         0.000000
```

```
[ ]: import seaborn as sns
plt.figure(figsize=(8,5))
sns.heatmap(reddit_cross, annot=True, cmap="Blues")
plt.title("Reddit: Topic-Sentiment Distribution")
plt.xlabel("Sentiment")
plt.ylabel("Topic")
plt.show()
```




```
[ ]: plt.figure(figsize=(8,5))
sns.heatmap(rappler_cross, annot=True, cmap="Greens")
plt.title("Rappler: Topic-Sentiment Distribution")
plt.xlabel("Sentiment")
plt.ylabel("Topic")
plt.show()
```

